

Example 1. Estimate each of the following limits graphically and numerically (use the given tables for your numerical estimates).

1. $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x}$

$x \rightarrow 0^-$	$\frac{\cos x - 1}{x}$	$x \rightarrow 0^+$	$\frac{\cos x - 1}{x}$
-.1		.1	
-.01		.01	
-.001		.001	
-.0001		.0001	

2. $\lim_{x \rightarrow 0} \sin \frac{1}{x}$

$x \rightarrow 0^-$	$\sin \frac{1}{x}$	$x \rightarrow 0^+$	$\sin \frac{1}{x}$
-.1		.1	
-.01		.01	
-.001		.001	
-.0001		.0001	
-.00001		.00001	

3. $\lim_{x \rightarrow 0} \frac{|x|}{x}$

$x \rightarrow 0^-$	$\frac{ x }{x}$	$x \rightarrow 0^+$	$\frac{ x }{x}$
-.1		.1	
-.01		.01	
-.001		.001	
-.0001		.0001	
-.00001		.00001	

4. $\lim_{x \rightarrow 0} \frac{1}{x}$

$x \rightarrow 0^-$	$\frac{1}{x}$	$x \rightarrow 0^+$	$\frac{1}{x}$
-.1		.1	
-.01		.01	
-.001		.001	
-.0001		.0001	
-.00001		.00001	